30 AI Advanced concepts for 2026



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Transformer Architecture

A deep learning architecture that uses self-attention to process sequences in parallel, replacing RNNs.

Self-Attention

Mechanism where each token attends to all other tokens to capture context.

Cross-Attention

Attention between two sequences (e.g., encoder output → decoder input).

Positional Encoding

Adds position information to tokens since transformers don’t inherently understand order.

Tokenization (BPE/WordPiece)

Splitting text into subword tokens for efficient vocabulary handling.

Context Window

Maximum number of tokens a model can process at once.

Attention Head

Multiple attention mechanisms running in parallel to capture different relationships.

Scaling Laws

Performance improves with more data, parameters, and compute.

Instruction Tuning

Fine-tuning models to follow human instructions better.

RLHF (Reinforcement Learning from Human Feedback)

Training method using human feedback to align outputs with preferences.

Zero-shot Prompting

Model performs tasks without examples.

Few-shot Prompting

Providing a few examples to guide model output.

Chain-of-Thought (CoT)

Prompting method that encourages step-by-step reasoning.

System Prompt

Defines model behavior and rules in applications.

Temperature / Top-p / Top-k

Sampling methods controlling randomness and diversity.

Hallucination Reduction

Techniques to reduce incorrect outputs (e.g., RAG, grounding).

Prompt Injection

A security risk where input manipulates model behavior.

Jailbreaking

Forcing the model to bypass safety constraints.

Guardrails

Rules and filters to ensure safe and controlled outputs

Output Parsing

Structuring model output into usable formats (JSON, schema).

RAG (Retrieval-Augmented Generation)

Combines retrieval systems + LLM to improve factual accuracy.

Vector Embeddings

Numerical representations capturing semantic meaning.

Vector Database

Optimized storage for embeddings with fast similarity search.

Semantic Search

Search based on meaning, not just keywords.

Chunking

Splitting large documents into smaller pieces for processing.

Re-ranking

Reordering retrieved results based on relevance scores.

Latency Optimization

Reducing response time using caching, batching, etc.

Caching in LLMs

Storing previous responses to reduce computation cost.

Fine-tuning vs Prompting

Choosing between model retraining vs prompt design.

Evaluation Metrics for LLMs

Measuring performance using BLEU, ROUGE, human eval.

Learn the concepts, not just the terms that's what cracks AI interviews



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